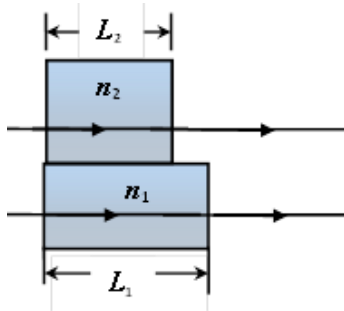


班级: ____ 姓名: ____ 学号: ____

Optics

1. Two waves of light in air, of wavelength $\lambda=600.0\text{nm}$, are initially in phase(同相位). They then travel through plastics layers as shown in Fig. with $L_1 = 4.00\ \mu\text{m}$, $L_2 = 3.50\ \mu\text{m}$, $n_1 = 1.40$, and $n_2 = 1.60$. (a) What the multiple of λ gives their optical path difference(光程差) after they both have emerged from the layers? (b) If the waves later arrive at some common point with the same amplitude, is their interference fully constructive, fully destructive, intermediate but closer to fully constructive, or intermediate but close to fully destructive? [(a)0.833, (b)more nearly constructive than destructive]



2. In a double-slit experiment it is found that blue light of wavelength 460nm gives a second-order maximum at a certain location on the screen. What wavelength of visible light would have a minimum at the same location? [613nm]

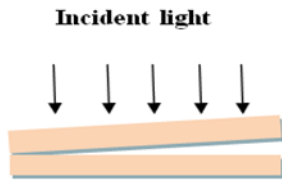
3. A very thin sheet of plastic ($n=1.6$) covers one slit of a double-slit apparatus illuminated by 640-nm light. The center point on the screen, instead of being a maximum, is dark. What is the (minimum) thickness of the plastic?[533nm]

4. A lens appears greenish yellow ($\lambda = 570\text{nm}$ is strongest) when white light reflects from it. What minimum thickness of coating ($n=1.28$) do you think is used on such a (glass) lens? [**223nm**]

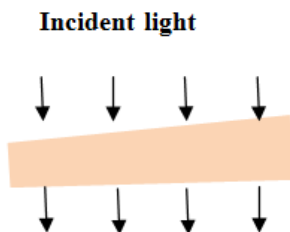
5. A thin film of alcohol ($n=1.36$) lies on a flat glass plate ($n=1.51$). When monochromatic light (单色光), whose wavelength can be changed, is incident normally, the reflected light is a minimum (光强极小) for $\lambda=512\text{nm}$ and a maximum for $\lambda=640\text{nm}$. What is the thickness of the film? [**471nm**]

6. A plane wave of monochromatic light is incident normally on a uniform thin film of oil that covers a glass plate. The wavelength of the source can be varied continuously. Fully destructive interference of the reflected light is observed for wavelengths of 500 and 700 nm and for no wavelengths in between. If the index of refraction of the oil is 1.30 and that of the glass is 1.50, find the thickness of the oil film. [**673nm**]

7. In Fig., a broad beam of wavelength 683nm is sent directly downward through the top plate of a pair of glass plates. The plates are 120 mm long, touch at the left end, and separated by $48\mu\text{m}$ at the right end. The air between the plates acts as a thin film. How many bright fringes will be seen by an observer looking down through the top plate?[141]

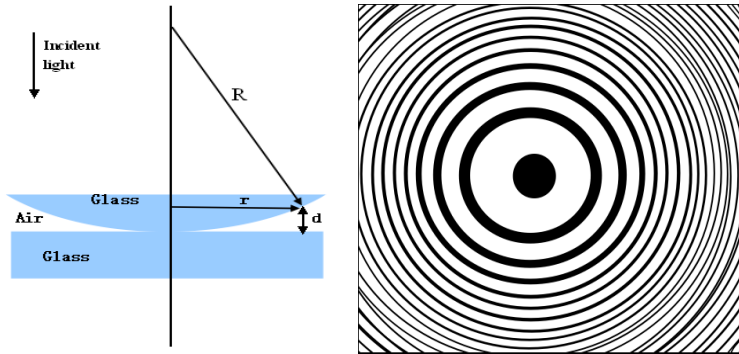


8. In Fig., a broad beam of wavelength 630 nm is incident at 90° on a thin, wedge-shaped film with index of refraction 1.50 . An observer intercepting the light transmitted by the film sees 10 bright and 9 dark fringes along the length of the film. By how much does the film thickness change over this length? [$1.89\mu\text{m}$]

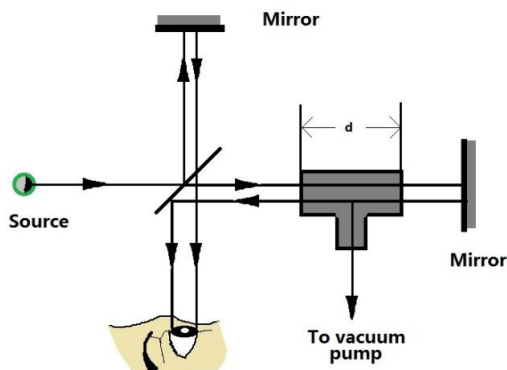


9. The figure shows a lens with radius of curvature R lying on a flat plate and illuminated from above by light with wavelength λ . Figure (a photograph taken from above the lens) shows that circular interference fringes (called *Newton's rings*) appear, associated with the variable thickness d of the air film between the lens and the plate. Find the radii r of the interference maxima assuming $r/R \ll 1$.

$$\left[r = \sqrt{\frac{(2m+1)R\lambda}{2}}. \quad m=0,1,2,\dots \quad \text{Or} \quad r = \sqrt{\frac{(2m-1)R\lambda}{2}}. \quad m=1,2,3,\dots \right]$$



10. In Fig. an airtight chamber of length $d=5.0\text{cm}$ is placed in one arm of a Michelson interferometer. (The glass window on each end of the chamber has negligible thickness.) Light of wavelength $\lambda=500\text{nm}$ is used. Evacuating the air from the chamber causes a shift of 60 bright fringes. From these data and to six significant figures (6 位有效数字), find the index of refraction (折射率) of air at atmospheric pressure. [1.00030]



11. Monochromatic light of wavelength 689 nm falls on a slit. If the angle between the first bright fringes on either side of the central maximum is 38° , estimate the slit width(狭缝宽度). [**3.2 μ m**]

12. When violet light of wavelength 415nm falls on a single slit, it creates a central diffraction peak(中央衍射峰) that is 9.20 cm wide on a screen that is 2.55 m away. How wide is the slit? [**23 μ m**]

13. How many fringes are contained in the central diffraction peak for a double-slit pattern (双缝衍射) if (a) $d=2.00a$, (b) $d=12.0a$, (c) $d=4.50a$, (d) $d=7.20a$. [(a) **3**, (b) **23**, (c) **9**, (d) **15**]

14. White light containing wavelengths from 400 nm to 750 nm falls on a grating with 7800 lines/cm. How wide is the first-order spectrum on a screen 2.80m away? [**1.10m**]

15. When yellow sodium light, $\lambda = 589 \text{ nm}$, falls on a diffraction grating, its first-order peak on a screen 60.0 cm away falls 3.32 cm from the central peak. Another source produces a line 3.71cm from the central peak. What is its wavelength? How many lines/cm are on the grating? [**658nm, 938lines/cm**]

16. A slit 1.00 mm wide is illuminated by light of wavelength 589nm. We see a diffraction pattern on a screen 3.00m away. What is the distance between the first two diffraction minima on the same side of the central diffraction maximum? [**1.77mm**]

17. Millimeter-wave radar (毫米波雷达) generates a narrower beam than conventional microwave radar, making it less vulnerable (易受攻击) to antiradar missiles than conventional radar.(a) Calculate the angular width (全角宽度) 2θ of the central maximum, from first minimum to first minimum, produced by a 220 GHz radar beam emitted by a 55.0-cm-diameter circular antenna. (b) What is 2θ for a more conventional circular antenna that has a diameter of 2.3m and emits at wavelength 1.6 cm?[(a) 6.04×10^{-3} rad (0.346°) (b) 1.7×10^{-2} rad (or 0.97°)]

18. A beam of light consisting of wavelengths from 460.0 nm to 640.0 nm is directed perpendicularly onto a diffraction grating with 160 lines/mm. (a) What is the lowest order that is overlapped by another order? (b) What is the highest order for which the complete wavelength range of the beam is present? In that highest order, at what angle does the light at wavelength (c) 460.0 nm and (d) 640.0 nm appear? (e) What is the greatest angle at which the light at wavelength 460.0 nm appears?

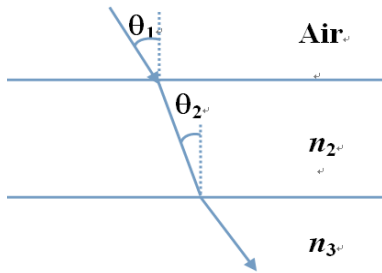
[(a) The first indication of overlap occurs when we compute the angle for $m = 4$ and $\lambda = 460$ nm; the result is 17.12° which clearly shows overlap with the large-wavelength portion of the $m = 3$ spectrum. (b) ninth order (c) 41.5° (d) 67.2° (e) 73.1°]

19. A diffraction grating having 180 lines/mm is illuminated with a light signal containing only two wavelengths, $\lambda_1 = 400 \text{ nm}$ and $\lambda_2 = 500 \text{ nm}$. The signal is incident perpendicularly on the grating. (a) What is the angular separation between the second-order maxima of these two wavelengths? (b) What is the smallest angle at which two of the resulting maxima are superimposed? (c) What is the highest order for which maxima for both wavelengths are present in the diffraction pattern?
[(a) 2.1° (b) 21° (c) 11]

20. At what angle should the axes of two polaroids(偏振片) be placed so as to reduce the intensity of the incident unpolarized light to (a) $\frac{1}{3}$, (b) $\frac{1}{10}$? **[(a) 35° (b) 63°]**

21. Two polarizers are oriented at 34.0° to one another. Light polarized at a 17.0° angle to each polarizer passes through both. What reduction in intensity takes place?
[37.1%]

- 22.** A light ray in air is incident on a flat layer of material 2 that has an index of refraction $n_2=1.5$. Beneath material 2 is in material 3 with an index of refraction n_3 . The ray is incident on the air material 2 interface at the Brewster angle for that interface. The ray of light refraction into material 3 happens to be incident on the material 2-material 3 interface at the Brewster angle for that interface. What is the value of n_3 ? **$[n_3 = 1.0]$**



- 23.** Unpolarized light with an intensity 25 W/m^2 is sent into a system of four polarizing sheets with polarizing directions at angles $\theta_1=40^\circ$, $\theta_2=20^\circ$, $\theta_3=20^\circ$, and $\theta_4=30^\circ$. What is the intensity of the light that emerges from the system? **$[0.50 \text{ W/m}^2]$**

